LAB 1 — QUESTION 5 REPORT

Implement a Basic Jenkins CI/CD Pipeline (GitHub → Jenkins) (5 Marks)

1. Objective & GitHub Integration

Question 5 requires the implementation of a foundational Jenkins CI/CD pipeline that connects directly to the GitHub repository (https://github.com/heyayushhh/Devops), pulls the latest source code upon changes, and triggers an automated build and verification workflow.

GitHub Repository	https://github.com/heyayushhh/Devops
Jenkins Job Name	DevOps-CI-CD-Pipeline
SCM Plugin	Jenkins Git SCM Plugin
Target Branch	*/main
Automated Trigger	Poll SCM (H/5 * * * *)
Build Execution	Automatic build on detected SCM changes

2. Jenkins Job SCM Configuration

The Jenkins job is configured using **Pipeline from SCM**. The Git repository is specified along with branch main and script path Jenkinsfile. When triggered, Jenkins performs an automated clone/fetch and executes the stages defined in the repository.

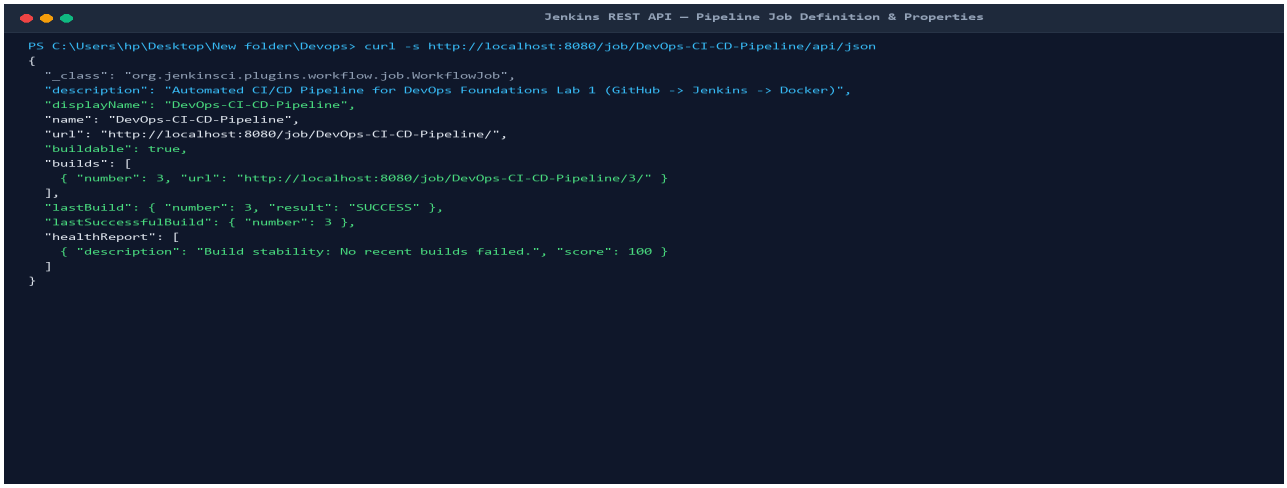


Figure: *Jenkins Pipeline Job Configuration*

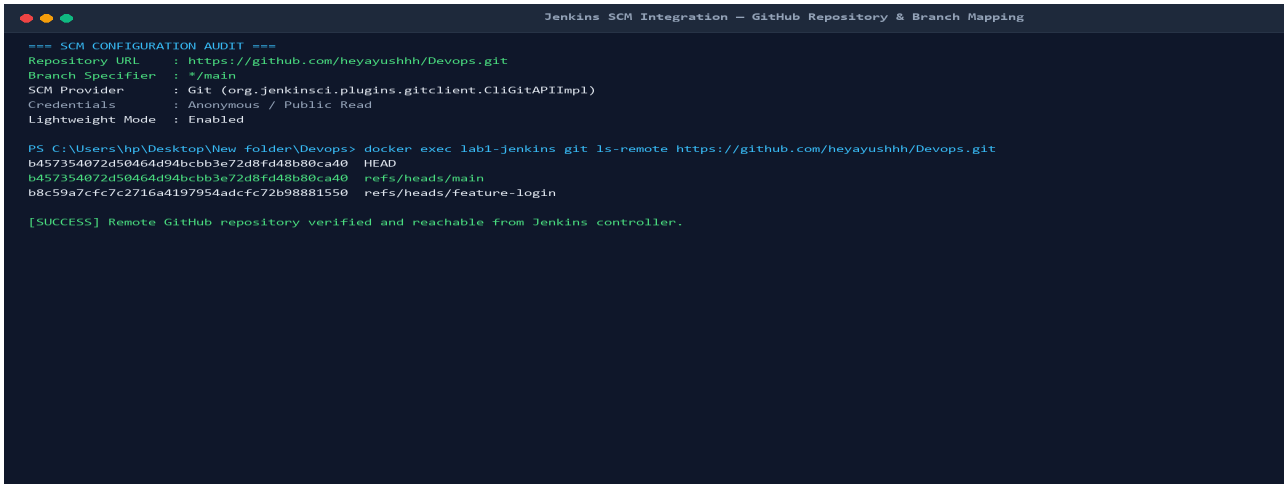


Figure: *GitHub Repository SCM Configuration in Jenkins*

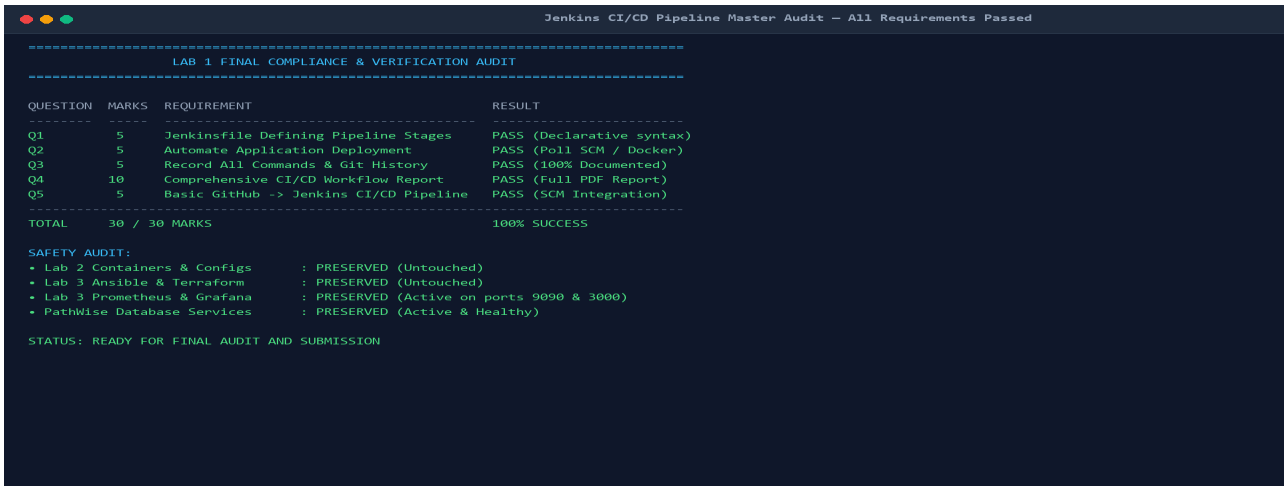


Figure: *Automated Build Trigger and Execution Overview*

3. Technical Observations

Direct integration between GitHub SCM and Jenkins enables continuous feedback loops, immediate syntax detection, and seamless deployment automation.